Predicting Interstitial ELF and Mapping Void Connectivity for Electride Screening

Electron localization function (ELF) prediction with an atom–void graph network, and exact geometric void connectivity

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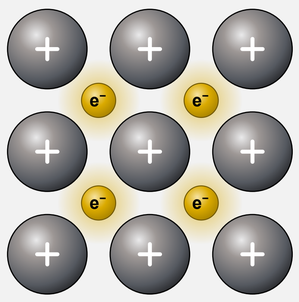
LOGO

QR CODE

BACKGROUND

What is an electride?

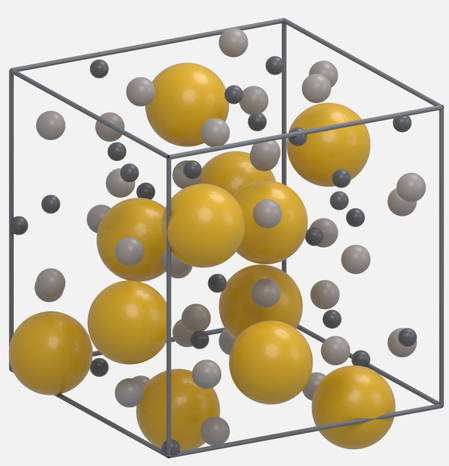
In an electride, extra electrons sit in the empty space between atoms and act as anions. Whether those regions are isolated (0D) or connected (1D, 2D) shapes the material's properties.



ELF measures how localized electrons are compared with a uniform electron gas. In electrides, an ELF maximum in an interstitial void can signal localized interstitial electrons. Computing ELF requires a DFT calculation.

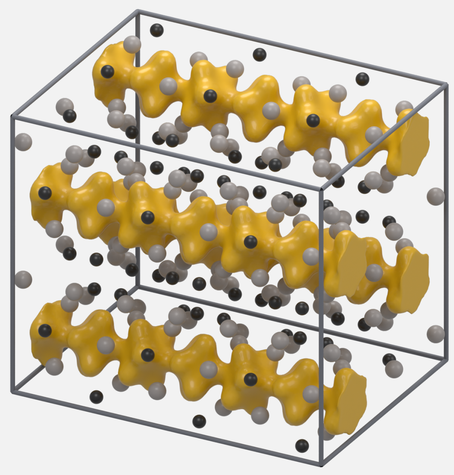
Electron connectivity sets the dimensionality

0D Isolated pockets



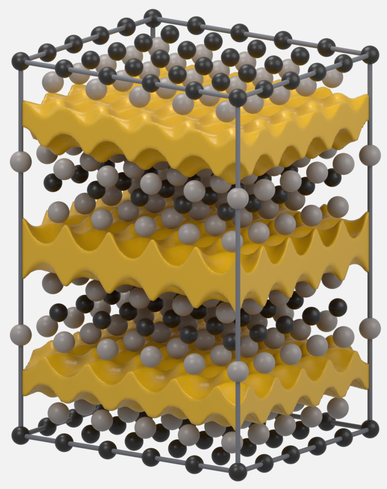
Mayenite-type $Ca_6Al_7O_{16}$: separate cages

1D Channels



Y_5Si_3: channels along one direction

2D Sheets



Ca_2N: layers between the atoms

GOAL

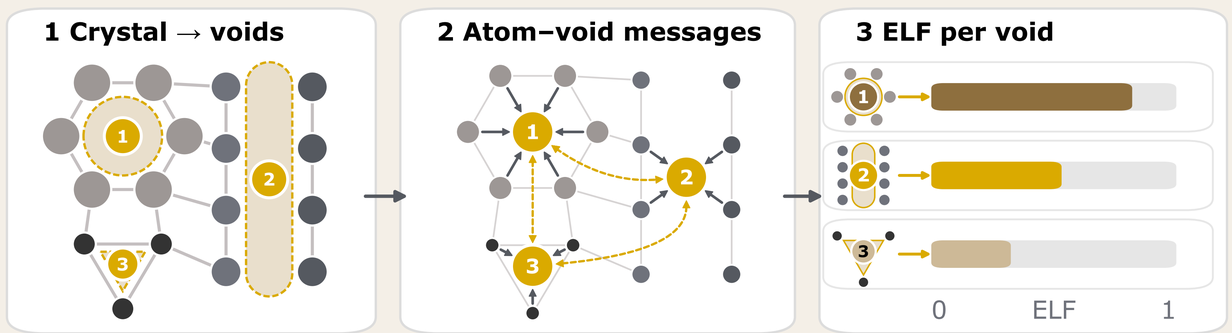
Screen for electrides from structure alone

- **No DFT:** predict ELF from atom positions.
- **Per void, not per crystal:** shows where electrons sit and how they could connect.
- **Two tools,** tested separately below.

TOOL 1 · PREDICT ELF

Can a graph network predict ELF at each candidate void from the surrounding atoms?

One ELF prediction per void, using only its local atomic region

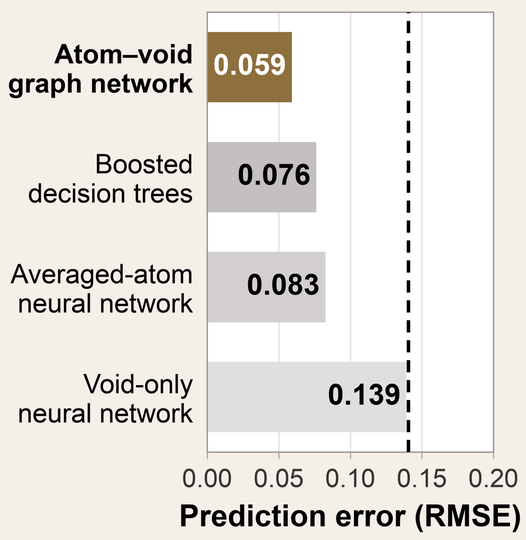


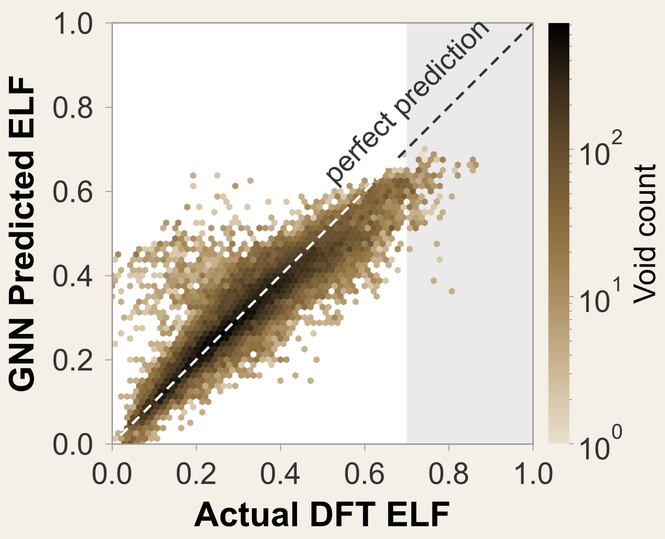
Test-set performance

88,680 test voids, 5,000 crystals

0.059 RMSE (ELF)

0.82 R^2





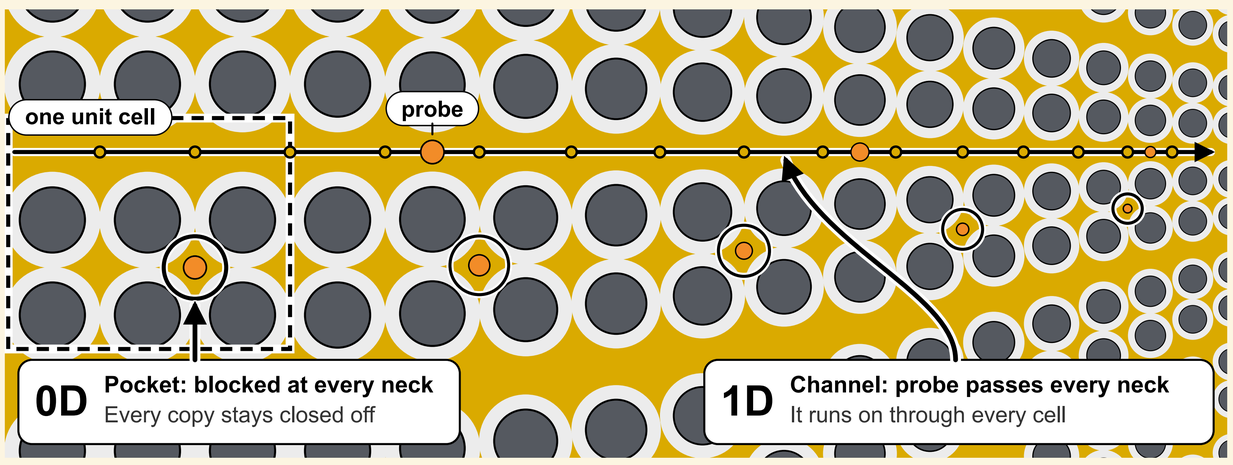
TOOL 2 · MAP HOW VOIDS CONNECT

Can geometry alone recover the dimensionality of known electrides?

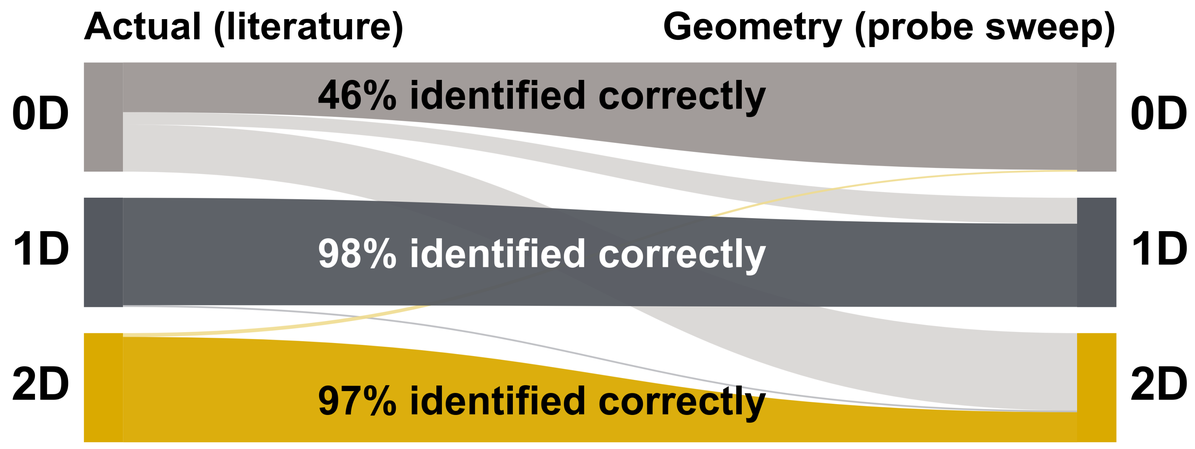
Probe must fit to pass through a neck

Atom + probe radius blocked: probe center cannot enter

Open space where the probe can sit



Predicted vs actual void connectivity

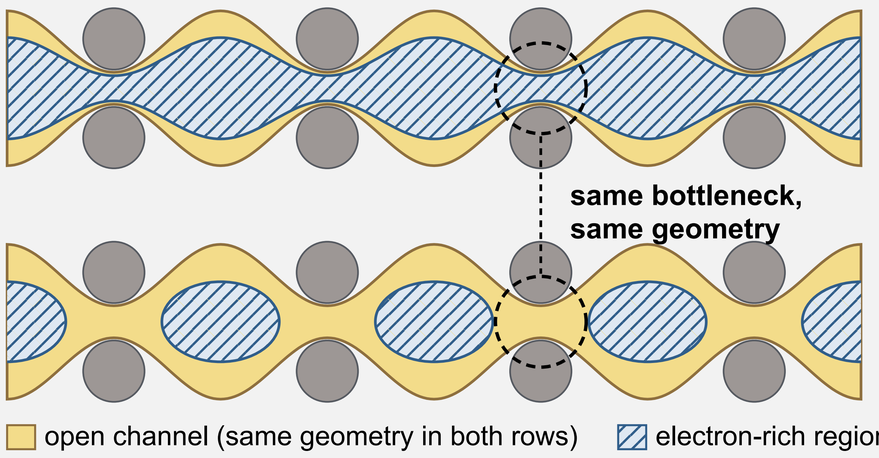


Many 0D electrides have connected open space

PROPOSED COMBINATION · FUTURE WORK

Geometry alone can't predict ELF connectivity

TOOL 2 · GEOMETRY — Find the channel and its bottleneck ▶ TOOL 1 · GNN — Predict ELF at that bottleneck ▶ COMBINED — Bottleneck ELF decides 1D vs. 0D



Connected electron-rich region

High ELF at the bottleneck
→ electrons connect (1D)

Separate electron-rich pockets

Low ELF at the bottleneck
→ isolated pockets (0D)

open channel (same geometry in both rows) · electron-rich region (high ELF) · atom

TAKEAWAY

- **Geometry shows where a path could exist**
- **ELF shows whether electrons occupy it**
- **Together they separate structural from electronic dimensionality**

NEXT STEPS

- Predict ELF at bottlenecks, not just voids
- Validate against DFT on known electrides
- Combine both into one electride screen

References [1] Becke & Edgecombe, J. Chem. Phys. 92, 5397 (1990), ELF. [2] Ellis, Ellis & Miao, J. Mater. Inf. 6, 45 (2026), doi:10.20517/jmi.2026.44, ELFNet. [3] Registry labels: Magnetic electrides, J. Am. Chem. Soc. (2023), doi:10.1021/jacs.3c00284; Zhu, Frolov & Choudhary, Matter 1, 1293 (2019); Burton et al., Chem. Mater. 30, 7521 (2018). [4] Methods: Shannon, Acta Cryst. A32, 751 (1976), ionic radii; Willems et al., Micropor. Mesopor. Mater. 149, 134 (2012), Zeo++; Gilmer et al., ICML (2017), message passing. **Acknowledgements:** [advisor, funding]; Purdue RCAC (Negishi).